

Ganesh Gupta

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EXPERIENCE

QureAI | ML Infrastructure Engineer

Mar 2026 – Present

- Raised **top-3 retrieval recall from 61% to 89%** on a 2K-query eval set by rebuilding the pipeline with **hybrid dense-sparse retrieval** combining semantic similarity and lexical relevance, plus **re-ranking**.
- Designed a **Neo4j** knowledge graph for **graph-aware retrieval** in production, modeling domain entities and relationships to improve result relevance.
- Rebuilt a synchronous retrieval service into an **async, autoscaled** architecture on **GCP Cloud Run**, cutting **p99 latency from 420ms to 140ms** across **50K+ daily requests**.

Oracle | Software Engineer, AI Platform

Jun 2025 – Present

- Engineered a **distributed executor** routing pipeline nodes across partitioned compute via typed **REST APIs**, backed by an **S3** content-addressed **artifact store**, sustaining **sub-200ms p99 latency**.
- Wrote **Go/Rust** scheduling and **fault-tolerant placement** logic for a multi-cloud **GPU/HPC fleet**, maintaining utilization through node loss and rebalancing under capacity constraints.
- Implemented **OAuth**-based auth and **RBAC**; participated in **design reviews** and **code reviews** on distributed service architecture.

Nomura Research Institute Financial Technologies | Software Engineer

Sep 2019 – Aug 2023

- Architected and operated **25+ production Java microservices** for **1M+ users** at **99.9% uptime** over 4 years on **AWS (EC2, RDS, SQS)**, owning **system design** end-to-end.
- Designed high-throughput **SQL** and **NoSQL** data pipelines, cutting nightly aggregation from **8 minutes to 90 seconds** via **indexed views** and batch partitioning.
- Led **architecture and code-review** sessions, mentoring engineers on **Object-Oriented Design** and distributed-systems reliability.

PROJECTS

Distributed KV Store

github.com/ganeshhgupta/Distributed-KV-Store

- Built a replicated key-value store in **Go** implementing **Raft consensus**, **linearizable reads**, and **log compaction**, tolerating node failures without data loss.

LangFetch

tinyurl.com/LangFetch

- Built an **agentic SQL** workspace that decomposes multi-table questions into dependency-ordered subqueries with intermediate result passing over **PostgreSQL**.

Watch-your-LLM

github.com/ganeshhgupta/watch-your-llm

- Built a self-hosted observability platform tracing requests, **latency**, and cost through a **FastAPI** collector and **PostgreSQL** store.

EDUCATION

University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0*

Aug 2023 – May 2025

MCKV Institute of Engineering | *B.Tech Computer Science*

Aug 2016 – Jul 2020

SKILLS

Languages: Java, Go, Python, Rust, C++, SQL

Retrieval & Search: Hybrid Dense-Sparse Retrieval, Re-Ranking, Neo4j Graph Retrieval, Relevance Evaluation

Distributed Systems: Object-Oriented Design, Raft Consensus, Fault-Tolerant Placement, Partitioned Compute

Cloud & Infra: AWS (EC2, RDS, SQS), GCP, Docker, CI/CD

Certifications: AWS Certified Developer Associate, GCP Professional MLOps Engineer